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SPECIAL ISSUE ARTICLE

Uncommon penalties for common problems

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Penalties are great tools to steer parametric and nonparametric models in desired directions. Familiar examples are the ridge and least absolute shrinkage and selection operator penalty to stabilize regression and roughness penalties to fit smooth trends. The purpose of this paper is to showcase some uncommon extensions to these penalties that help to solve common problems. Switching from the usual sum of squares in the penalty to sums of smaller powers can achieve very sparse results. Asymmetric penalties make it easy to enforce shape constraints. In some cases it is effective to work with a sum of 2 signals, each having its own "tailored" penalty to adapt it to specific properties of the data. Many examples, in 1 and 2 dimensions, mostly based on real data, illustrate the (moderate) theory.

KEYWORDS

L_p norm, ridge regression, shape constraints

1 | INTRODUCTION

Many chemometricians are familiar with penalties. The prototypical example is ridge regression,[1,2] which was quickly adopted by the chemometrics community. At its time of publication, ridge regression was a revolutionary break with the obsession for unbiased estimation that dominated statistics. Hoerl and Kennard showed that by constraining regression coefficients desirable results could be obtained.

Penalties also occur in smoothing contexts. Whenever parameters of a model have a natural order, one can constrain the differences between neighbors and in this way tune their smoothness. This is the roughness penalty in action.[3,4] Actually this type of penalty predates ridge regression: it was already proposed by Whittaker,[5] who used it to smooth life tables.

In this paper I hope to show that many penalty variations are possible, which can help to extract desirable and meaningful results from data. Not all examples come from chemometrics. If an application in another field is very convincing I use it. In most cases I show graphs, so it will be easy for the reader to imagine application of the ideas presented here to his or her own data.

I do not indicate explicitly whether a penalty is common or not; it is a rather subjective label anyway. Also, this paper is not meant to be a review of penalties. It is more a kind of appetizer, showing the potential of tailored penalties in a variety of problems.

A few examples were computed with MATLAB, the rest with R,[6] a free and powerful computing environment. I will be happy to share the programs that produced the graphs in this paper.

The plan of the paper is as follows. Section 2 presents basic ideas and an overview of the key points of this paper: How to combine a penalty and a measure of fit of a model to data. A penalty forces a model, gently or not, in a desired direction. The fit can be measured by a sum of squares or a log-likelihood. Almost any penalty can be combined with any measure of fit, which is of great practical value.

Section 3 is the central part of the paper. It presents a tour of many different penalties, common and uncommon, and shows their use on (mostly real) data. When presenting different penalties, I skip some computational details; they are the subject of Section 4. Many penalties, especially those for smoothness and shape, have multidimensional counterparts; this is the subject of Section 5. With each penalty comes a tuning parameter, to set its influence. Generally, results depend heavily on the value of that parameter. It is desirable to let it be determined by the data. Section 6 discusses what is possible and what is not. The Discussion section closes the paper.

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2 | THE BASIC IDEAS

Suppose we fit a statistical model to data. The observations are given as a vector y and the model is $\mu(\beta)$, where β are the parameters and μ gives the expected values of y. To estimate β, an objective or goal function is optimized:

$$\textbf{Goal} = \textbf{Fit}(y; \mu(\beta)).$$

Here **Fit** measures the distance between the data and the model. Often it is the sum of the squares of the residuals, the differences between y and μ. In other cases, it can be (minus) a log-likelihood, when the distribution of y is patently non-normal, as for counts or binary data. In robust statistics the sum of absolute differences is a popular choice. Note that $\mu(\beta)$ does not have to be linear. In fact, in some applications, a strongly nonlinear function, like the exponential, is used.

Minimizing the goal in Equation 2 gives the model all freedom to get as close to the data as possible. This is desirable when the number of parameters is relatively small and there are many observations. Then we are in the arena of classical statistics. Unfortunately, as models become larger (having more parameters), disadvantages appear. The fit to the data becomes better, and might even get perfect, but the uncertainty in the parameter estimates increases. In linear regression this is a well known phenomenon, but it occurs in many other settings.

Large uncertainties in parameter estimates are a problem when interpreting a model, because confidence intervals will be wide. It is an even greater problem when a model is used for prediction, because uncertainty intervals will be (very) wide too.

The classical solution for dealing with (too) many explanatory variables is to apply selection, finding a subset that strikes a middle ground, and combining a good fit with not too large standard errors of the parameter estimates. Several selection procedures have been developed over the years, but their popularity is slowly declining.

A more attractive solution is to introduce a penalty. The goal is extended to:

$$\textbf{Goal} = \textbf{Fit}(y; \mu(\beta)) + \textbf{Penalty}(\beta) = F(y; \mu(\beta)) + \lambda P(\beta). \quad (1)$$

The penalty is a nonnegative function $P(.)$ of only the parameters and it is added to the function $F(.)$ that measures the fit to the data. The (positive) parameter λ sets the balance between fit and penalty. Actually the fit will get worse, so what do we gain? A well chosen penalty constrains the parameter vector in a desirable way. But what is desirable? It depends on our goals. In ridge regression we constrain the size of the parameters, reducing standard errors and improving prediction performance. In trend estimation, we compute a $\mu(\beta)$ that is smooth but keeps close to a given signal.

Because we have a sum of 2 functions in Equation 1, we can adapt both of them independently to our goals. The whole range of classical statistical (regression) models can be called upon, with proper distributions and a proper transformation between parameters and μ. The penalty can be chosen independently to influence size, shape, or roughness of the solution.

An extension of Equation 1 is a sum of multiple penalties, a perfect tool when we have several constraints on a model. For every extra condition a proper penalty term is added in Equation 1. An example is ridge regression under the condition that all coefficients have to be nonnegative.

A further extension is to let μ be a sum of multiple components, each with a "tailored" penalty. An example is a piecewise constant signal on top of a smooth trend. Using the right penalties, the components can be separated almost magically.

3 | A PENALTY TOUR

3.1 | Ridge regression and variations

Assume we are given a response vector y and a matrix of predictors X and that the model $y = X\beta + e$ is a reasonable one. Here e is the noise, having a symmetric distribution with constant variance. The goal is to estimate β by least squares.

This is a standard problem in statistics and $\hat{\beta} = (X'X)^{-1}X'y$ gives the solution explicitly. If X has a handful of columns and the number of observations is not small, this is straightforward. But as the number of columns increases, the results become more and more sensitive to small changes in the data and at a certain point the problem even becomes singular. This is known as multicollinearity and the problem is called ill-conditioned.

Hoerl and Kennard[2] proposed ridge regression as a remedy. The objective function is:

$$S_2 = \sum_i (y_i - \sum_j x_{ij}\beta_j)^2 + \kappa \sum_j \beta_j^2 = \|(y - X\beta)\|_2^2 + \kappa\|\beta\|_2^2. \quad (2)$$

The first term is the familiar sum of squares measure of fit and the second term the penalty, which constrains the size of β. This is called shrinking. When $\kappa = 0$, we have standard linear regression. Ridge regression has an explicit solution:

$$\hat{\beta} = (X'X + \kappa I)^{-1}X'y, \quad (3)$$

where I is the identity matrix.

The goal of a ridge penalty is modest: It only constrains the size of the parameter vector. The LASSO[7] is more aggressive. The abbreviation stands for least absolute shrinkage and selection operator and indicates what it does: to shrink parameter values and to select a subset of them. Those not selected are set to 0. In situations where one hopes to separate the chaff from the wheat this is an attractive property. The idea is that by using the LASSO only the influential variables in a regression model are selected.

The LASSO replaces the sum of squares in the penalty by the sum of absolute values

$$S_1 = \sum_i (y_i - \sum_j x_{ij}\beta_j)^2 + \kappa \sum_j |\beta_j| = \|(y - X\beta)\|_2^2 + \kappa\|\beta\|_1. \quad (4)$$

No explicit solution is available. For a given κ one can use a quadratic programming algorithm. In the statistical community so-called path algorithms are more popular. Least angle regression is very similar to the LASSO.[8] Starting from a very large value, where all elements of $\hat{\beta}$ are equal to 0, κ is gradually reduced. The traces of the individual regression coefficients can be informative. An alternative, iterative algorithm is presented in Section 4.

A useful notation is $\sum_j |\beta_j|^p = L_p(\beta)$. In loose mathematical jargon: the L_p sum of β. It is often called the L_p norm, but officially the norm is $(\sum_j |\beta_j|^p)^{1/p}$. I will call it the L_p penalty. Ridge regression uses $p = 2$ and LASSO $p = 1$. Both choices guarantee that the objective function (S_2 or S_1) is convex, meaning that there is a unique global minimum and that a good minimization algorithm will move downhill to find it. Convexity is lost when we take $p < 1$, so there might be multiple local minima of:

$$S_p = \sum_i (y_i - \sum_j x_{ij}\beta_j)^2 + \kappa \sum_j |\beta_j|^p. \quad (5)$$

Frank and Friedman[9] coined the name bridge penalty for $p < 1$. Again no explicit solution is available and there is no guarantee that when a numerical procedure finds a minimum it is the global one. Yet the bridge penalty, especially with $p = 0$, is attractive, because it is a very aggressive selector, much better than the LASSO, without shrinking. I illustrate that with an example of spike deconvolution. Computational details can be found in Section 4.

Assume that y is the (noisy) output of a linear system that received a sparse series of spikes as input. The goal is to estimate the positions and the sizes of the spikes, knowing the impulse response of the system.[10] The situation can be modeled by taking $y = X\beta + e$, where the columns of X contain shifted copies of the impulse response. In Figure 1 we see simulated data and the results of regression with $p = 2$, 1, and 0. It is clear that ridge regression is of no use here. LASSO works well, but it is biased: The spikes are too small and there is some "debris" at their feet, compensating for the inherent shrinkage of the spike. In this application the L_0 penalty does an excellent job. The fit to the data is very good and essentially the same for all choices of the penalty.

An iterative algorithm was used, which started from the L_1 solution; details can be found in the cited paper. The same approach has been used for large-scale image deconvolution

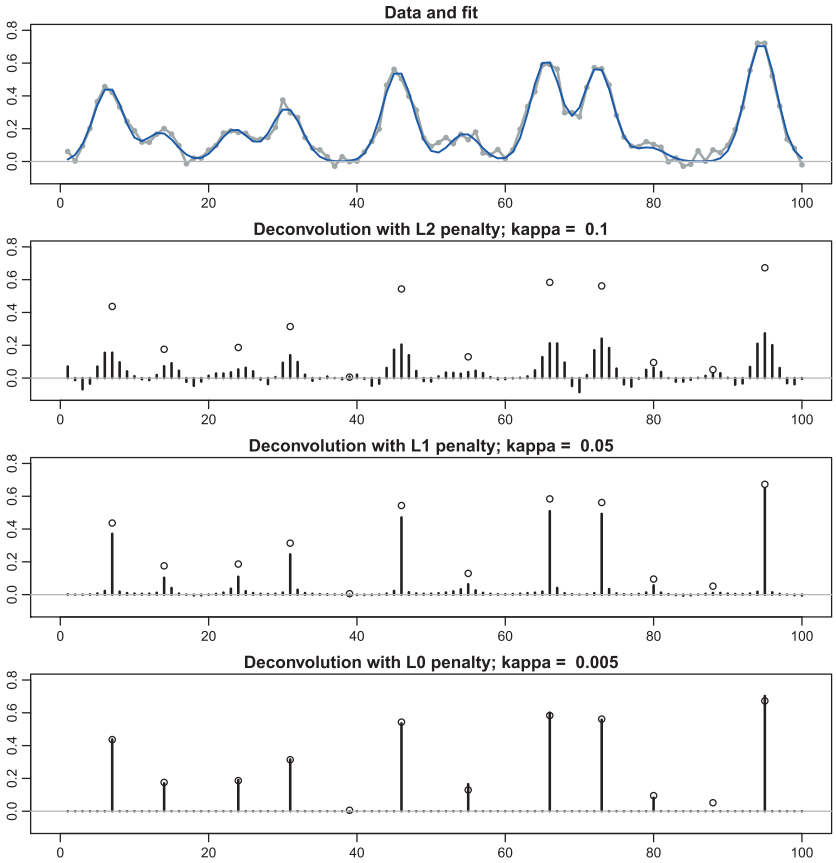


FIGURE 1 Penalized deconvolution of a system driven by spikes (simulated data). The grey connected dots in the top panel show the data, the blue curve the fit The other panels show, as vertical bars, the estimated input spikes obtained using different powers in the size penalty (as indicated in the titles of the panels). The small circles show the true values

in fluorescence nanomicroscopy.[11] A shape penalty was used there too, to force the spikes to be nonnegative. See Section 3.3 for details.

3.2 | Roughness penalties

Many analytical measurements have a natural order, along time, wavelength, chemical shift, or some other variable. A roughness penalty can be exploited in several ways: to reduce noise[4] or to regularize regression on ordered explanatory variables, as in multivariate calibration.[12]

3.2.1 | Whittaker smoothing and variants

There are several ways to measure the roughness of a vector μ. If the observations are placed on a uniform grid, differences between the values of neighboring elements are a natural choice. Introducing the operator notation $\Delta\mu_i = \mu_i - \mu_{i-1}$, the roughness can be measured as $R(\mu) = \sum^i(\Delta\mu_i)^2$, and $\lambda R(\mu)$ can be used as a penalty. Varying λ we can set the balance between fit and roughness to our taste. Higher order differences give smoother results. The notation for second differences is $\Delta^2\mu_i = \Delta\mu_i - \Delta\mu_{i-1}$. Higher order differences follow straightforwardly by repeating differencing: $\Delta^{d+1}\mu_i = \Delta^d\mu_i - \Delta^d\mu_{i-1}$.

The Whittaker smoother[4,5] minimizes the penalized sum of squares

$$S = \sum_i w_i(y_i - \mu_i)^2 + \lambda \sum_i (\Delta^d \mu_i)^2. \quad (6)$$

The weights w are useful when observations are missing. Setting $w_i = 0$ and giving the corresponding y_i an arbitrary value, interpolated (or extrapolated) values for $\hat{\mu}$ are calculated automatically. Weights are also needed when smoothing data for which a sum of squares is not appropriate as a measure of fit. We will see an example later on, when smoothing the counts in a histogram.

Minimization of S in 6 leads to the linear equations

$$(W + \lambda D'D)\hat{\mu} = Wy. \quad (7)$$

Here D is the matrix that forms differences of order d by multiplication: $D\mu = \Delta^d\mu$. The number of equations is equal to the number of observations, meaning that it can get large for long data series. However, the system is very sparse, because $D'D$ is banded, with $2d$ nonzero subdiagonals at each side of the main diagonal. Using sparse matrices, the time to solve the system grows only linearly with the length of y. In MATLAB only a few lines of code are needed:

```
n = length(y)
D = diff(speye(n), d)
W = spdiags(w, 0, n, n)
mu = (W + lambda * D ' * D)\ (w .* y)
```

It takes less than a second to smooth (with second order differences) a series of one million observations, on an average personal computer.

Figure 2 shows an example of Whittaker smoothing of noisy nuclear magnetic resonance (NMR) spectrum. The details in the lower panel shows that a peak is not fitted well if it is too sharp. This is a problem for any smoother. The fit of the peak can be improved by less smoothing, but then the result will be too wiggly in other parts of the spectrum.

In Equation 6 the measure of fit contains weights. They allow to regulate the influence of individual observations. Missing data are a special case: We set the missing elements of y to 0 and the corresponding weights too, and automatic interpolation will occur. For automatic extrapolation just extend the data with "missing observations" at one or both sides.

When λ is made very large and second order differences are used, the fitted curve approaches a straight line, which is actually the least squares regression line through the data. With third order differences a parabola is obtained in the limit.

Another type of limit, $\mu_i = a\sin(2\pi i/b + c)$, is obtained with $\mu_i - 2\phi\mu_{i-1} + \mu_{i-2}$ in the penalty instead of $\Delta^2\mu_i = \mu_i - 2\mu_{i-1} + \mu_{i-2}$, if $\phi = \cos(2\pi/b)$, where b is the period of the sine wave. This is useful when we are smoothing cyclic data of which the (approximate) period is known. An example (removal of 60 Hz power-line noise) is provided in Section 3.4.

A limitation of the Whittaker smoother is that the observations have to be on a uniform grid. For time series and spectra this usually is the case but not in a scatterplot of arbitrary (x, y) pairs. P-splines[3,13] solve this issue by inserting a basis of B-splines between the data and a vector of

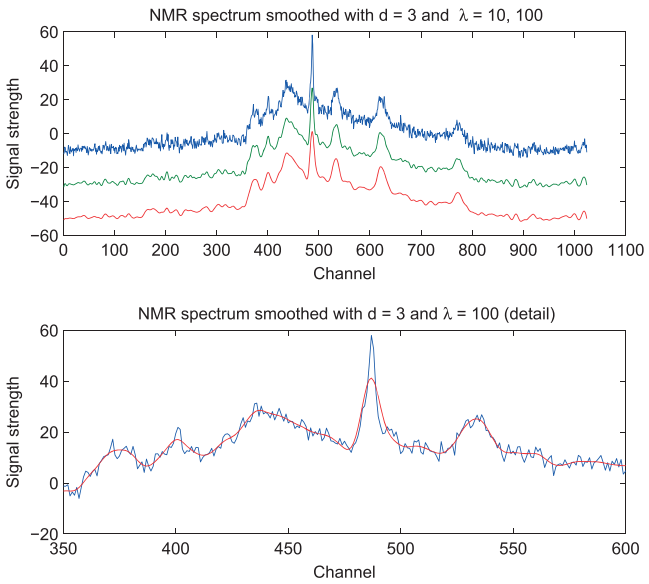


FIGURE 2 Whittaker smoothing of an NMR spectrum, using a third order difference penalty. Top panel: the raw spectrum and and the smooths for two values of λ (vertically shifted for clarity). Bottom panel: detail, showing data and smooth for $\lambda = 100$

coefficients: $\mu_i = \sum^j B_j(x_i)\beta_j$, or $\mu = B\beta$ with $B = [b_{ij}]$ and $b_{ij} = B_j(x_i)$. Without a penalty smoothing with B-splines boils down to linear regression. But the objective function of P-splines is

$$S = \|y - B\beta\|_2^2 + \lambda\|D\beta\|_2^2, \tag{8}$$

with D a matrix of the proper size that forms differences of order d.

Figure 3 shows P-splines in action. Each panel contains 28 (scaled) B-splines, in rainbow colors. The black dots are the (simulated) data and the grey line the P-spline fit, the sum of all B-splines. In the upper-left panel λ is small, the penalty is weak, and the heights of neighboring splines, which

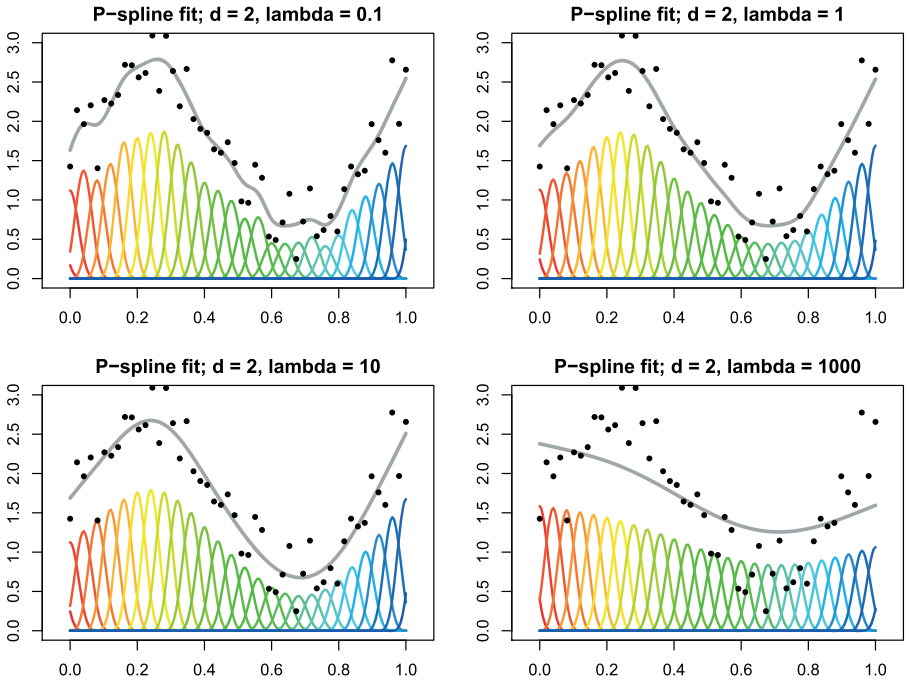


FIGURE 3 Illustration of P-splines. The B-splines in thr basis are drawn in rainbow colors. The black dots show the data and the grey line the fit, which is ths um of the B-splines. The values of the penalty parameter λ varies between the panels. When it is low the fit is wiggly, but it is smooth when λ is large

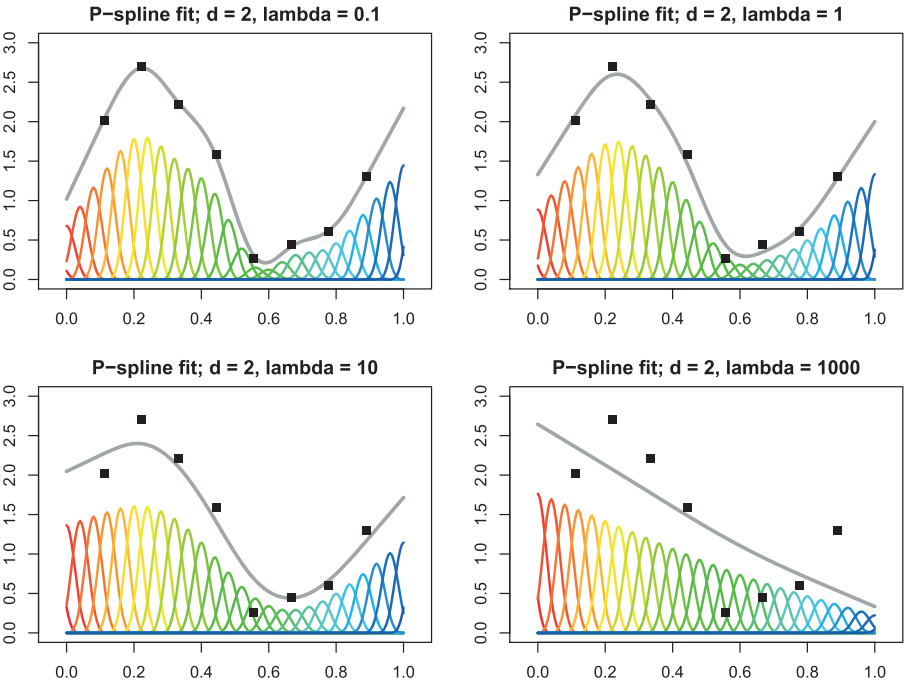


FIGURE 4 Illustration of interpolation and extrapolation with P-splines. Colored B-splines as in the previous figure. Although there are only 8 data points and 28 B-splines, no problems occur. Not the extrapolation at both ends

reflect their coefficients, vary relatively strongly. In the lower right panel λ is large and as a consequence, the heights of the B-splines vary gradually, resulting in a smooth fit.

Without a penalty, the smoothness of a B-spline fit is determined by the number of splines. But with a penalty smoothness can be tuned independently. Also it solves potential problems when data are missing in larger parts of the domain. In addition it allows extrapolation as illustrated in Figure 4.

3.2.2 | Histogram smoothing

To smooth a histogram containing counts y, it is assumed that the y_i was drawn from a Poisson distribution with expect value μ_i. In the objective function the Poisson log-likelihood and the difference penalty are combined. The fit of μ, the expected counts, is measured by minus two times the log-likelihood. The minus sign is to turn maximization into minimization and the factor 2 to simplify the resulting equations. Another change is that the penalty is not on μ itself, but on $\eta = \log\mu$. This is inspired by the theory of generalized linear models.[14] The objective function is

$$\begin{aligned} S &= -2\sum_i \log\left(\mu_i^{y_i} e^{-\mu_i}\right) + \lambda \sum_i \left(\Delta^d \eta_i\right)^2 \\ &= -2\sum_i \left(y_i \eta_i - e^{\eta_i}\right) + \lambda \sum_i \left(\Delta^d \eta_i\right)^2. \end{aligned} \tag{9}$$

After setting the derivatives with respect to the elements of η equal to 0, we obtain the penalized likelihood equations

$$\lambda D' D\eta = y - \mu, \tag{10}$$

with D again the matrix that forms d-th order differences. This is a nonlinear equation, because $\mu = e^{\eta}$. To solve it iteratively by linearization, write $\eta = \tilde{\eta} + (\eta - \tilde{\eta})$, where $\tilde{\eta}$ is an approximation to the solution. Then

$$\mu_i = \exp(\tilde{\eta}_i + (\eta_i - \tilde{\eta}_i)) = \exp(\tilde{\eta}_i)\exp(\eta_i - \tilde{\eta}_i) = \tilde{\mu}_i \exp(\eta_i - \tilde{\eta}_i).$$

It is known that $e^u \approx 1 + u$ if u is a number close to 0. This means that $\exp(\eta_i - \tilde{\eta}_i) \approx 1 + \eta_i - \tilde{\eta}_i$ and $\mu_i \approx \tilde{\mu}_i + \tilde{\mu}_i(\eta_i - \tilde{\eta}_i)$, or in matrix notation $\mu \approx \tilde{\mu} + \tilde{M}(\eta - \tilde{\eta})$, where $\tilde{M} = \mathrm{diag}(\tilde{\mu})$. Inserting this result into Equation 10 we get the linear system

$$(\tilde{M} + \lambda D' D)\eta = y - \tilde{\mu} + \tilde{M}\tilde{\eta}. \tag{11}$$

Starting from $\tilde{\eta} = \log(y+1)$, convergence is quick: 5 or 6 iterations are generally sufficient for 6-digit accuracy. Note that Equation 11 is a case of the Whittaker smoother in Equation 7, with a special right-hand side.

Figure 5 shows an example in the top panel. The data are torsion angles of the protein *2bfu*, obtained from the PDB database (www.rcsb.org/pdb). A nice smooth curve is obtained, but there are problems at the edges. Angles are circular and need special treatment, but the penalty does not account for that. This is solved by adding extra terms to the sum of squared differences in the penalty, to connect left and right ends of the domain. With a second order penalty, the

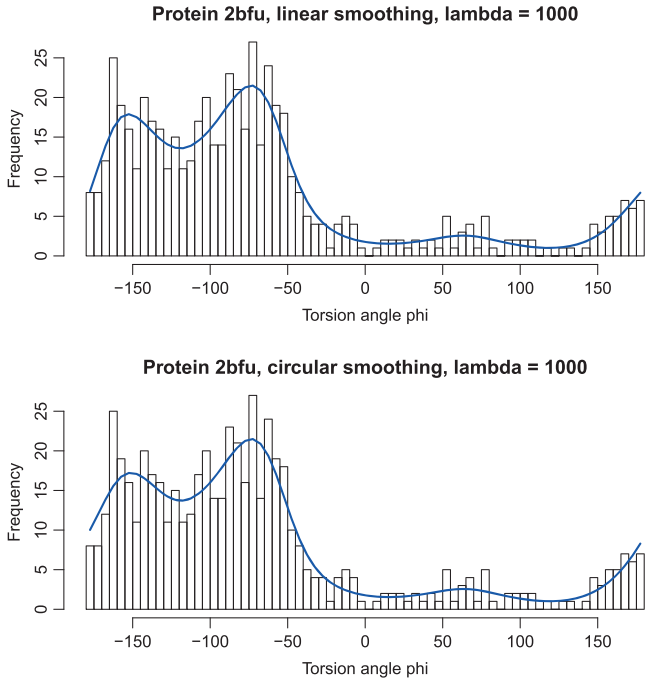


FIGURE 5 Smoothing of a histogram of torsion angles of a protein. Top: with a standard penalty on third differences. Note that the data are circular, but that start and end of the curve do not connect smoothly. Bottom: with a penalty that accounts for the circular nature of the data, having three extra terms that connect start and end

2 extra terms are $(\eta_1 - 2\eta_n + \eta_{n-1})^2$ and $(\eta_2 - 2\eta_1 + \eta_n)^2$, where n is the length of y. In a similar way, for a third order penalty 3 extra boundary bridging terms are added, each on 4 elements of η. Now left and right ends connect smoothly, as the bottom panel of Figure 5 shows.

3.2.3 | Segmented smoothing

The roughness penalty in the Whittaker smoother uses a sum of squares of differences, but the L_1 or L_0 sum can be applied too. With first order differences the fit to the data will be piecewise constant, because many differences will be forced to be 0. With second order differences a piecewise linear fit will be obtained.

Figure 6 shows an application to copy number variations in tumor DNA.[15] Normal DNA has 2 copies of each chromosome, but in tumors that number can locally be larger (amplifications) or smaller (deletions). The L_0 penalty on differences leads to a piecewise constant fit. To find the solution, the L_0 sum of differences is approximated by a weighted sum of squares, as detailed in Section 4. Frommlet and Nuel[16] discuss L_0 penalties in detail.

Geyer[17] presented data on the occurrence of Down syndrome in newborn babies in British Columbia from 1961 to 1970, stratified for the age of the mother. Figure 7 shows that after age 30 the probability of occurrence rises sharply. A plot of the log-odds suggest a piecewise linear curve for $\eta = \log(p/(1-p))$, with p as the probability. A generalized linear model is based on the binomial log-likelihood,

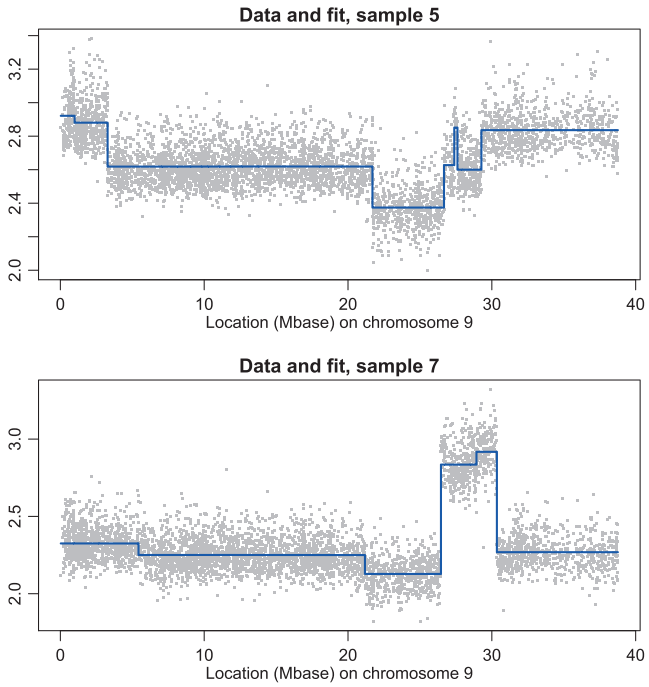


FIGURE 6 Smoothing of DNA copy number data using an L_0 penalty on differences. This penalty favors a piecewise-constant fit. The vertical axes are not calibrated: offsets and scales are arbitrary

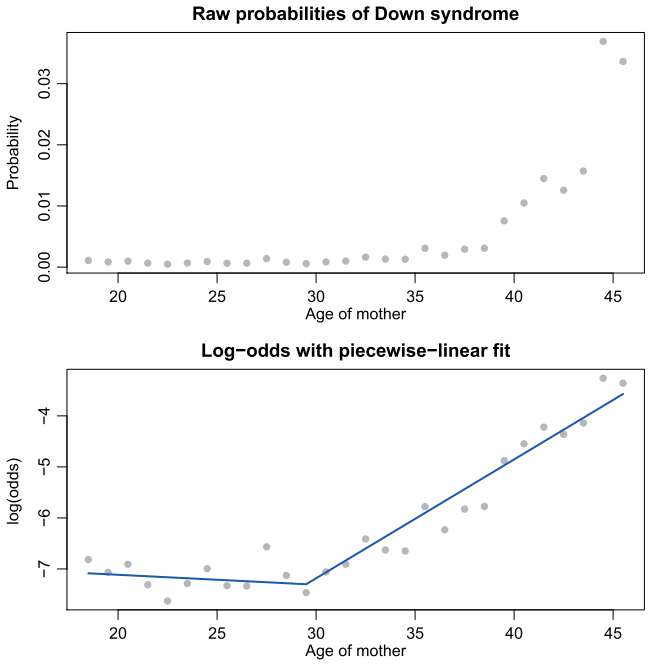


FIGURE 7 The probability of getting a baby with Down syndrome given the age of the mother. Top: on the probability scale. Bottom: on the log-probability scale, with a piecewise-linear fit obtained with a generalized linear model and an L_0 penalty on second order differences

$\sum_i(y_i \log p_i + (t_i - y_i)\log(1 - p_i))$, where t_i is the number of births at age i and y_i the number of Down cases. In a similar way as for histogram smoothing, penalized likelihood equations follow, which can be linearized and solved iteratively. I skip the details.

An L_0 penalty on second differences of η aims for a piecewise linear fit. The result, with $\kappa = 10$, is shown in Figure 7.

3.3 | Shape penalties

Well-chosen penalties can force the fit of a (smooth) model to have a desired shape. Differences of a certain order can (locally) be forced to be 0, or to a have a desired sign, positive or negative. The former case is easily handled with an extra penalty, while the latter case calls for an asymmetrically weighted penalty. Examples will follow.

Some shape constraints ask for a locally flat part of the fitted curve. The fit to the Down syndrome data in Figure 7 provides an example. The initial segment is sloping slightly downward. It seems reasonable to assume that it is flat. This can be achieved by adding the penalty $\kappa(\eta_2 - \eta_1)^2$, with κ a large number. In fact we can choose any pair of adjacent elements of η on the initial segment, with identical results. If we have a strong opinion about the slope, say that it should be s, the penalty can be changed to $\kappa(\eta_2 - \eta_1 - s)^2$.

Figure 8 shows a similar constraint in action. The data are the daily flows that a gas distribution station delivered to households at different temperatures.[18] There are no observations below -5 centigrade. It is of interest to know how much can be delivered at lower temperatures. A naive approach is to fit P-splines and use them to extrapolate outside the data. The broken red curve shows the result, which is optimistic and unrealistic. From the physical background it is known that the capacity of the station is limited and so the flow should level off at very low temperatures. This can be achieved with a heavy penalty on the first differences of the 3 leftmost spline coefficients, resulting in the blue curve. In a similar way a constant level at high temperatures can be forced.

When the constraint is on the signs of differences we need to switch to asymmetric penalties. Consider the following

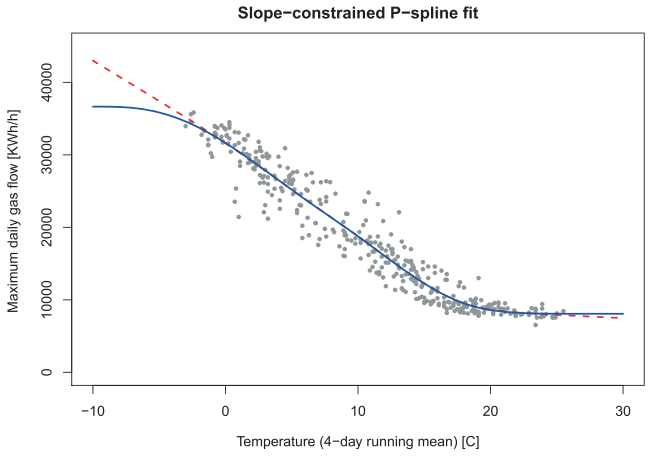


FIGURE 8 Gas flow versus temperature in a distribution station. The dots show the observations. The broken red line is a P-spline fit without constraints. The full blue line shows the fit that is forced to have zero derivatives at both ends

extension of ridge regression with an extra penalty, as third term, in the objective function:

$$S = \sum_i (y_i - \sum_j x_{ij}\beta_j)^2 + \kappa \sum_j \beta_j^2 + \psi \sum_j v_j \beta_j^2, \quad (12)$$

where the weights v are such that $v_j = 1$ if $\beta_j \leqslant 0$ and $v_j = 0$ if $\beta_j > 0$, and ψ is a large number. A concise notation uses the indicator function: $v_j = I(\beta_j \leqslant 0)$. Negative elements of β are heavily penalized, while positive elements are not. Some elements of $\hat{\beta}$ will be (very) small negative numbers, while the remaining ones are zero or positive. This is good approximation to a solution with only non-negative elements, especially if ψ is large.

The generalized ridge solution is given by

$$(X'X + \kappa I + \psi V)\hat{\beta} = X'y, \quad (13)$$

with $V = \text{diag}(v)$. We have a chicken-and-egg problem here : to compute $\hat{\beta}$ we need to know v, but to compute v we need to know $\hat{\beta}$. Practice has shown that an iterative solution works well: start with $v \equiv 0$, compute $\hat{\beta}$, update v, and repeat the recipe until convergence is obtained.

This idea can be extended to other shape constraints using asymmetrically weighted differences in an extra penalty. With $\psi \sum_j I(\Delta\beta_j < 0)(\Delta\beta_j)^2$ a monotone increase is obtained. In a smoothing context the objective function will be

$$S = \sum_i w_i(y_i - \mu_i)^2 + \lambda \sum_i (\Delta^2 \mu_i)^2 + \psi \sum_i I(\Delta\mu_i < 0)(\Delta\mu_i)^2. \quad (14)$$

To get a monotone decrease, $\psi I(\Delta\mu_i < 0)(\Delta\mu_i)^2$ should be used.

To enforce a convex or concave shape, second order differences should be used in the asymmetric shape penalty. Multiple penalties can also be combined to get, say, a positive decreasing solution. Shape penalties can also be combined with an L_0 penalty on the size of parameters. Deconvolution provides an example: in some applications, like image deconvolution,[11] it is known that the input spikes are positive and their estimates should reflect that.

Returning to ridge regression with positive parameters, we have to realize that it is not strictly true that no negative values will occur. In fact, for every parameter for which $v_j = 1$, a small negative value will be obtained. Taking ψ large, these negative values can be made very small, which will be all right for most practical purposes. One can refine the recipe by setting a threshold at some small positive value instead of zero. Some experimenting will generally be needed to determine a proper threshold.

In principle, implementing shape constraints as penalties creates the possibility of balancing the constraint against the fit by playing with ψ. Probably this is only of theoretical value.

Sometimes it is useful to combine a transformation of μ with a shape penalty. An example is unimodal smoothing with

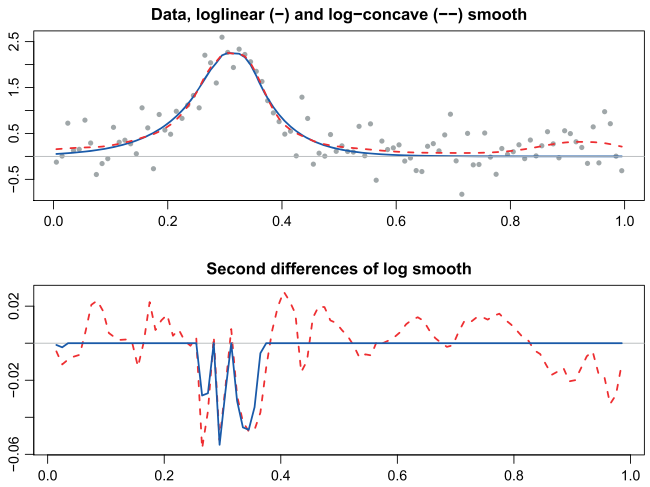


FIGURE 9 Log-concave unimodal smoothing: with (full blue lines) and without the asymmetric shape penalty (red broken lines)l $\lambda = 10$ and $\kappa = 10^6$. Top panel: data (simulated) and fitted curves. Bottom panel: second order differences of the logarithms of the fitted curves

$\eta = \log\mu$ and a penalty to make η concave.[19] The objective function of this log-concave model is

$$S = \sum_i (y_i - e^{\eta_i})^2 + \lambda \sum_i (\Delta^2 \eta_i)^2 + \psi \sum_i I(\Delta^2 \eta_i > 0)(\Delta^2 \eta_i)^2. \quad (15)$$

Here we have a standard penalty on second order differences, for smoothness, and an asymmetric one to force the log of the fit to be concave. The problem is non-linear and an solution can be found by linearization of the exponential function, as was done for histogram smoothing in Section 3.2.

The logarithm is crucial here, forcing μ itself to be concave would lead to completely different results. Figure 9 shows an example, with and without the shape penalty. Even without the extra penalty a quite good unimodal fit is obtained. But the lower panel shows that positive values of $\Delta^2\mu$ occur, which are suppressed if the shape penalty is added.

3.4 | Signal decomposition

Some signals can be decomposed in two (or more) components with properties that play into the hands of different "tailored" penalties. Two examples follow. One is the removal of 60 Hz power-line interference from a electrocardiogram and the other is separating a signal into steps and a smooth trend.

Figure 10 shows a small section of a distorted electrocardiogram. We can reduce the interference with the Whittaker smoother, but, as the figure shows, to really get rid of it quite strong smoothing is needed, rounding the peak in the data. An alternative is to explicitly estimate the two components. We know the frequency of the disturbance so we can specify a proper harmonic penalty with a heavy weight. For the other component we use the penalty on second order differences, with a relatively small weight. The objective function is

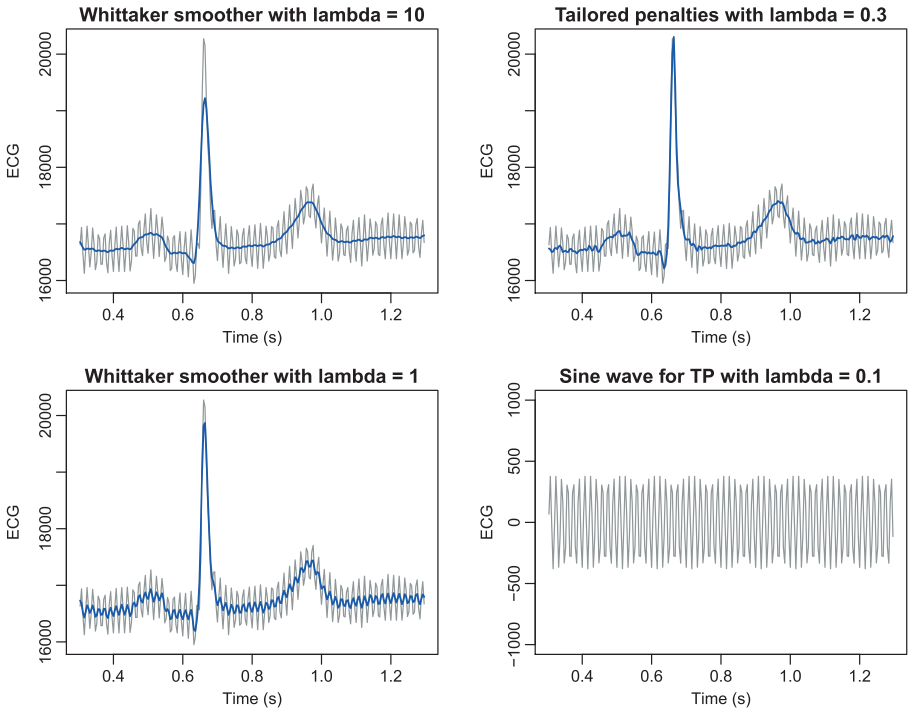


FIGURE 10 A short section of an ECG signal with 60 Hz interference. Top left: signal and the result of Whittaker smoothing with $\lambda = 10$ (data in grey, fit in blue); the interference is essentially removed, but the peak is rounded. Bottom left: Whittaker smoothing with $\lambda = 1$, leaving the peak almost intact, but keeping a part of the interference. Top right: ECG signal with and without interference removed, using the two-component model ($\lambda = 0.1$ and $\kappa = 10^5$). Bottom right: the estimated interference component. The apparently non-constant amplitude is a visual artifact, caused by the limited sampling frequency (250 Hz)

$$S = \sum_i (y_i - \mu_i - g_i)^2 + \lambda \sum_i \left(\Delta^2 \mu_i\right)^2 + \kappa \sum_i (g_{i-1} - 2\phi g_i + g_{i+1})^2, \tag{16}$$

where $\phi = \cos(2\pi f \Delta t)$, with $f = 60$, the frequency in Hertz, and $\Delta t = 0.004$ the distance between the samples in seconds. As can be seen in Figure 10, the improvement is large. The parameter values are $\lambda = 0.1$ and $\kappa = 10^5$.

It is remarkable that we can extract two meaningful components from one observed signal. It is a beautiful example of how properly chosen penalties can force a model towards desirable results. In fact we can hardly speak of a model: it is just the sum of two components, ECG signal and interference. This aspect of penalties has not been explored much.[20]

Figure 11 shows a piecewise-constant signal on top of a smooth trend. To fit the data, the objective function is

$$S = \sum_i (y_i - \mu_i - s_i)^2 + \lambda \sum_i \left(\Delta^2 \mu_i\right)^2 + \kappa \sum_i |s_i - s_{i-1}|^0, \tag{17}$$

where μ models the trend, having the standard roughness penalty and s models the steps, with an L_0 penalty on its first differences, as for the DNA copy number data in Figure 6.

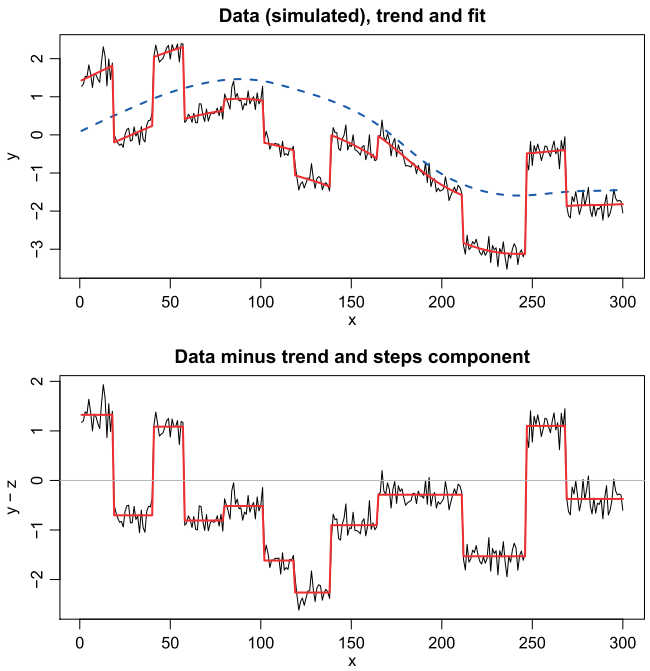


FIGURE 11 Simulated jumps on a trend. Top: the data, the fit and the estimated trend component (broken line). Bottom: de-trended data and fit

4 | COMPUTATIONAL DETAILS

The purpose of this paper is to give a taste of familiar and unfamiliar penalties. There is no room for detailed technical discussions. To maintain a pleasant flow, I have avoided to discuss several computational details when presenting and illustrating penalties. For some of them, like in ridge regression, implementation is almost trivial. In other cases, extra care pays off, in computation time and memory use.

The Whittaker smoother leads to a system of equations with a very small bandwidth. If sparse matrices are used, the time needed for storing and solving the equations grows linearly with the length of the data series.[4] For smoothing of large data sets P-splines are attractive, because the size of the system of equations is equal to the number of B-spline coefficients, usually much smaller than the number of observations.[3]

To implement the L_p sum with $p < 2$, iterated weighted squares are attractive. It is a generalization of a proposal[21] for curve fitting with the L_1 sum as the measure of fit. The idea is very simple. Suppose the penalty is $\kappa \sum_j |u_j|^p$, where each u_j can be a regression parameter, or a (higher order) difference. Now it is obvious that

$$|u_j|^p = |u_j|^2 / |u_j|^{2-p} = v_j |u_j|^2 \quad \text{with} \quad v_j = |u_j|^{p-2}. \tag{18}$$

This is a weighted square and the penalty can be approximated by $\kappa \sum_j v_j u_j^2$. To avoid numerical instabilities when u_j gets close to zero, a simple safety device is to use $v_j = 1/\sqrt{(u_j^2 + \gamma^2)}$ when $p = 1$ and $v_j = 1/(u_j^2 + \gamma^2)$ when $p = 0$. Here γ is a small number of the order of 10^{-5} times the expected maximum of $|u|$.

The trick is to fill in an approximation $\tilde{u}$ to get approximate weights $\tilde{v}$ and solve the problem with the easy to handle weighted quadratic penalty. Hopefully this leads to a better approximation to the solution and the process is repeated as long as is needed. In practice this works surprisingly well. However, convergence can be slow, 100 or more iterations may be needed, especially with the L_1 sum. On the other hand, it is almost trivial to implement arbitrary L_p sums this way and it can be combined with any measure of fit to the data. It also allows us to use sparse matrices without restrictions.

5 | TWO-DIMENSIONAL PENALTIES

Many of the penalties we encountered in the previous sections can be transplanted to two-dimensional problems on a grid. The data form a matrix $Y = [y_{ij}]$ and we fit the matrix $M = [\mu_{ij}]$. A simple approach is to smooth each column of Y separately and to smooth the rows of the resulting matrix. This gives a very fast and effective procedure. But there is a drawback: different weights for individual cells of Y are not possible. These are needed for generalized linear smoothing, for robust fitting, and for quantile or expectile smoothing.

A more general approach is to introduce an objective function, with weights in the matrix W, including penalties on every row and column of Z:

$$\begin{aligned} S = &\sum_i \sum_j w_{ij}(y_{ij} - \mu_{ij})^2 \\ &+ \lambda \sum_j \sum_i (\Delta^d \mu_{ij})^2 + \check{\lambda} \sum_i \sum_j (\check{\Delta}^d \mu_{ij})^2. \end{aligned} \tag{19}$$

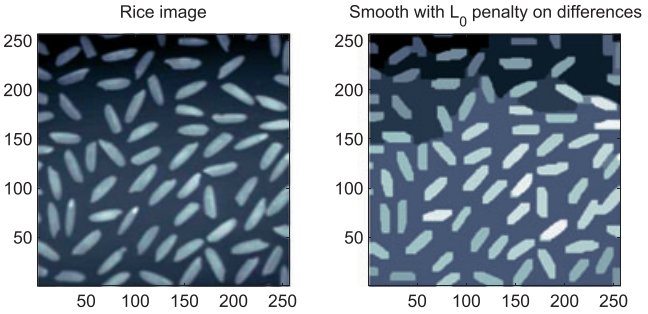


FIGURE 12 Segmentation of an image of rice grains using L_0 penalties on first differences of rows and columns ($\lambda = 0.005$)

Here the operator Δ forms differences down the columns of M and $\check{\Delta}$ does the same along its rows. The notation $\|.\|_2^2$, indicates sum of the squares of all elements of a matrix:

$$S = \|W \odot (Y - M)\|_2^2 + \lambda \|DM\|_2^2 + \check{\lambda} \|M\check{D}'\|_2^2, \tag{20}$$

where $W \odot X$ stands for the element-by-element product of W and X. The matrices D and $\check{D}$ form differences of order d. Note that λ and $\check{\lambda}$ can be different, allowing anisotropic smoothing.

The objective function in (19) is written in terms of the matrices W, Y and M. To compute a solution with matrix operations we have to turn them into vectors w, y and μ. Then we have to solve, with $W^* = \text{diag}(w)$ and $\otimes$ indicating the Kronecker product,

$$(W^* + \check{\lambda}(\check{D}'\check{D}) \otimes I + \lambda \check{I} \otimes D'D)\hat{\mu} = W^* y. \tag{21}$$

Here I ($\check{I}$) is an identity matrix with n ($\check{n}$) rows and columns if Y has n rows and $\check{n}$ columns, and $W^* = \text{diag}(w)$. The system is large ($n\check{n}$ by $n\check{n}$) and sparse, but with a relatively large bandwidth.

Unless sharp features, like jumps or kinks, have to modeled, it is a good idea to switch to tensor product P-splines.[22] The key idea is to write $M = BA\check{B}'$, where B and $\check{B}$ are B-spline bases along the rows and columns of Y, and to put the penalties on A. The latter matrix is much smaller than Z, say 20 rows and columns, leading to a relatively small system of equations to solve.

If needed, shape penalties can be added for rows, columns or both, also working on the elements of A. This way a smooth surface was fitted to psychological test data with a monotone increase along both dimensions.[23]

When we are interested in segmentation of an image, L_0 penalties on first differences of rows and columns work well. Figure 12 shows an example: an image of rice grains. Because sharp boundaries are wanted, it is not attractive to use splines here. Despite the large size of the system of equations (250^2 unknowns), computation (in Matlab 2012) is surprisingly fast. It took a little under 20 seconds to obtain the segmentation, in 87 iterations. The computer was an average desktop model with an Intel I3 processor.

The algorithm is robust against random noise, as Figure 13 shows. I consider these results only as a proof of principle and an illustration of an uncommon penalty. Further research is

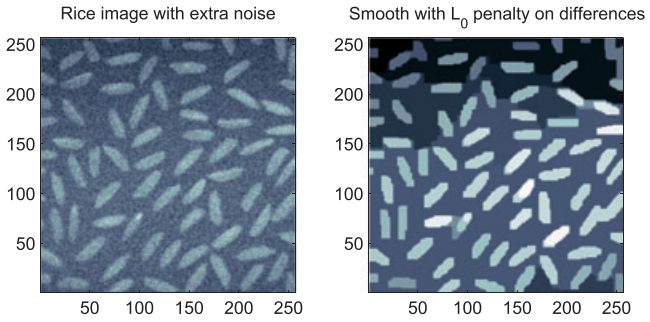


FIGURE 13 Segmentation of the image of rice grains, using L_0 penalties on first differences of rows and columns ($\lambda = 0.005$). Pseudo-random noise added

needed to determine whether it can compete with established image segmentation algorithms.

The left panel in Figure 12 shows that the background of the image has a trend, going from dark in the top left to brighter in the bottom right. This suggests a two-dimensional variant of the model with constant segments on top of a smooth trend (see Figure 11). To improve the efficiency of the computations, the trend can be modeled with tensor product P-splines, as it is expected to be very smooth.

6 | OPTIMAL PENALTIES?

The previous sections have shown that many types of penalties are possible and that they can serve us well. The weight of a penalty can be tuned to make a model fit to the data well, while conforming to our prior opinions like smoothness or sparseness. In some cases it is possible to find a good value for a penalty parameter in an automatic way. For ridge regression and standard smoothing, cross-validation and AIC have been explored extensively. The results can be satisfying.

When smoothing data to fit a trend, it is often assumed (implicitly) that noise is independent. If that is not the case under-smoothing can occur, because the serial correlation is confounded with the trend. In principle it is possible to model the noise by, say, an autoregressive model but this complicates the computations enormously, especially when we have to deal with non-normal data. A variant of the L-curve appears be attractive.[24]

In my examples I tuned the penalty parameters by eye, to get a pleasing result. One can call this subjective, and indeed it is. But the practicing chemometrician sometimes has to accept making dirty hands by using methods that are not completely objective. Generally (s)he will know enough about the data under study to make informed choices.

7 | DISCUSSION

The statistical models I presented consist of two parts: a regression component and one or more penalties. In many cases the regression part is trivial, like in the Whittaker smoother where we regress on the identity matrix. In fact the penalties are the more interesting component: they introduce sparseness or smoothness or enforce shape constraints.

The norm in the penalty is crucial. For ridge regression or simple smoothing the sum of squares (of differences) is appropriate. The L_1 or L_0 sum works better if sparseness is wanted. The latter gives by far the best results, because it does not shrink like the former. However, it leads to a non-convex optimization problem. The iterative fitting algorithm described in Section 4 is easy to implement and works well, so it is not hard to obtain a solution. But there is no guarantee that the global optimum has been found. In practice this sounds more dramatic than it is. The applications I described allow an easy visual check on the results. If a solution looks good, why worry too much?

Transformations play an important role in parts of my story. But they are not applied to the data but to the fit. In histogram smoothing we model a smooth logarithm of the expected counts. To get a unimodal fit we also work on the logarithm of the fitted curve, because with a proper shape penalty we can force it to be concave. This approach is familiar in statistics, where the generalized linear model is the prime example. Unfortunately it is rather unfamiliar in chemometric circles. Nevertheless I hope this will change.

I limited the use of asymmetry to penalties, but it can also be of values in the measure of fit. Quantile curves are can be obtained by combining asymmetric weights with absolute values of residuals. This is known as quantile regression or quantile smoothing.[25] Asymmetry can also be combined with a sume of squares of residuals; this has been used for baseline estimation.[26]

Is it art or science? I imagine that the critical reader will not find much of a guiding principle here for selecting penalties or models. Many of the examples I presented were products of a "creative flash", not of systematic research. Nevertheless I hope that many colleagues will make themselves familiar with penalties, common and uncommon, and come up with new ideas. May this paper be a source of inspiration.

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